University of Sheffield

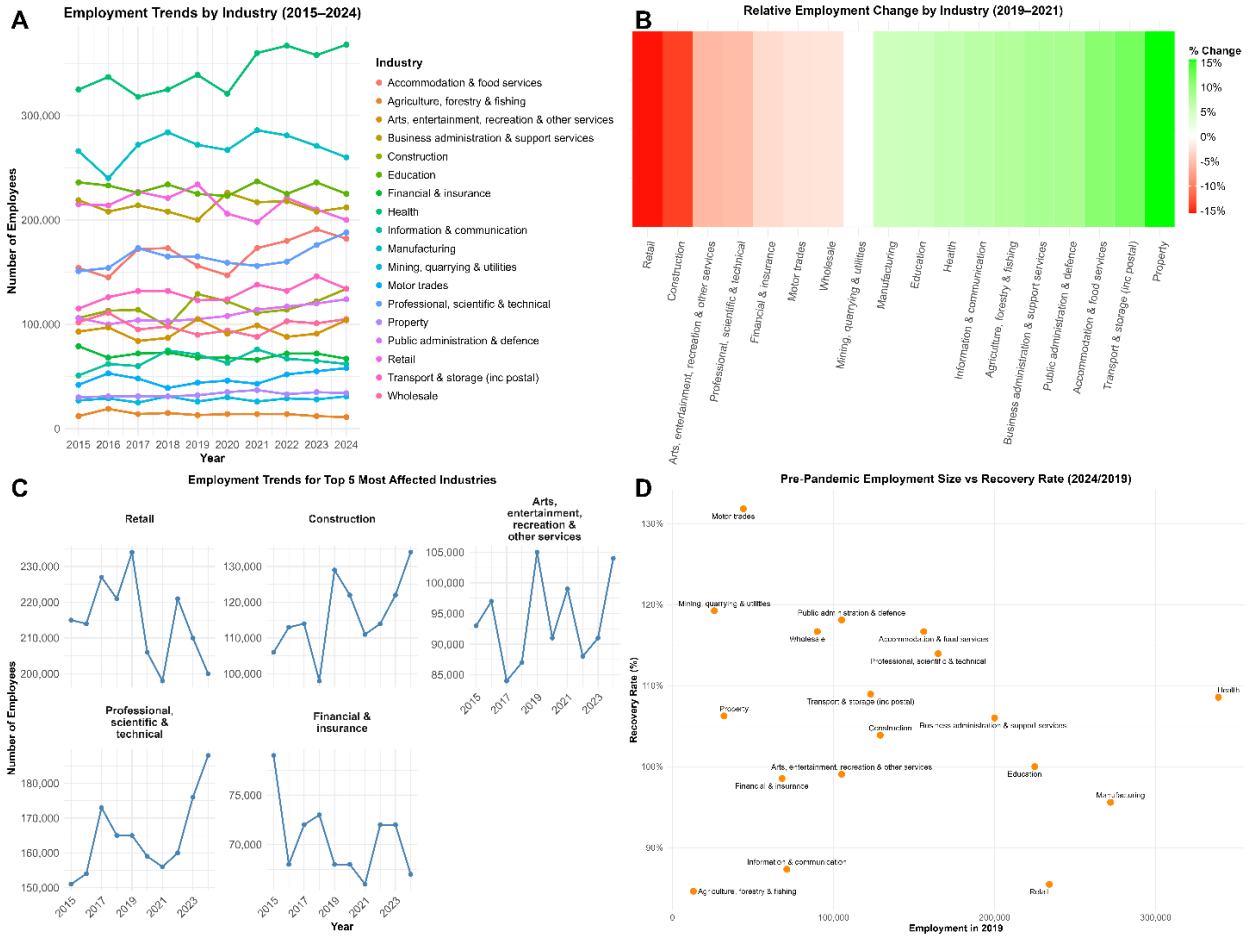
# Visualising Post-COVID Structural Changes in Yorkshire and the Humber Industry Employment (2015-2024)

UCHE DUMZO-AJUFO

# 1 COMPOSITE VISUALISATION

**Figure 1**

*Composite Visualisation of Employment Trends in Yorkshire and the Humber (2015 to 2024)*




*Note*. The composite visualisation combines four charts (A: line chart, B: heatmap, C: small multiples and D: scatterplot) to provide a multi-dimensional view of industry employment trends. It highlights temporal patterns, the immediate impact of COVID-19, recovery trajectories and relative resilience across sectors. Zoomed-in versions of each individual chart appear in section 2.1 of the report, allowing closer inspection of trends that may be difficult to discern in the smaller composite display.

# 2 MAIN REPORT

## 2.1 KNOWLEDGE BUILDING

The COVID-19 pandemic caused major disruptions to employment across the UK, with regional variations reflecting differences in industrial composition and resilience (Office for National Statistics, 2022). This section explores the structural changes in industry employment in Yorkshire and the Humber from 2015 to 2024, focusing on the pandemic's immediate impacts and subsequent recovery. The guiding research question is:

"What structural changes occurred in industry employment in Yorkshire and the Humber following the COVID-19 pandemic, and how did these changes vary across sectors?"

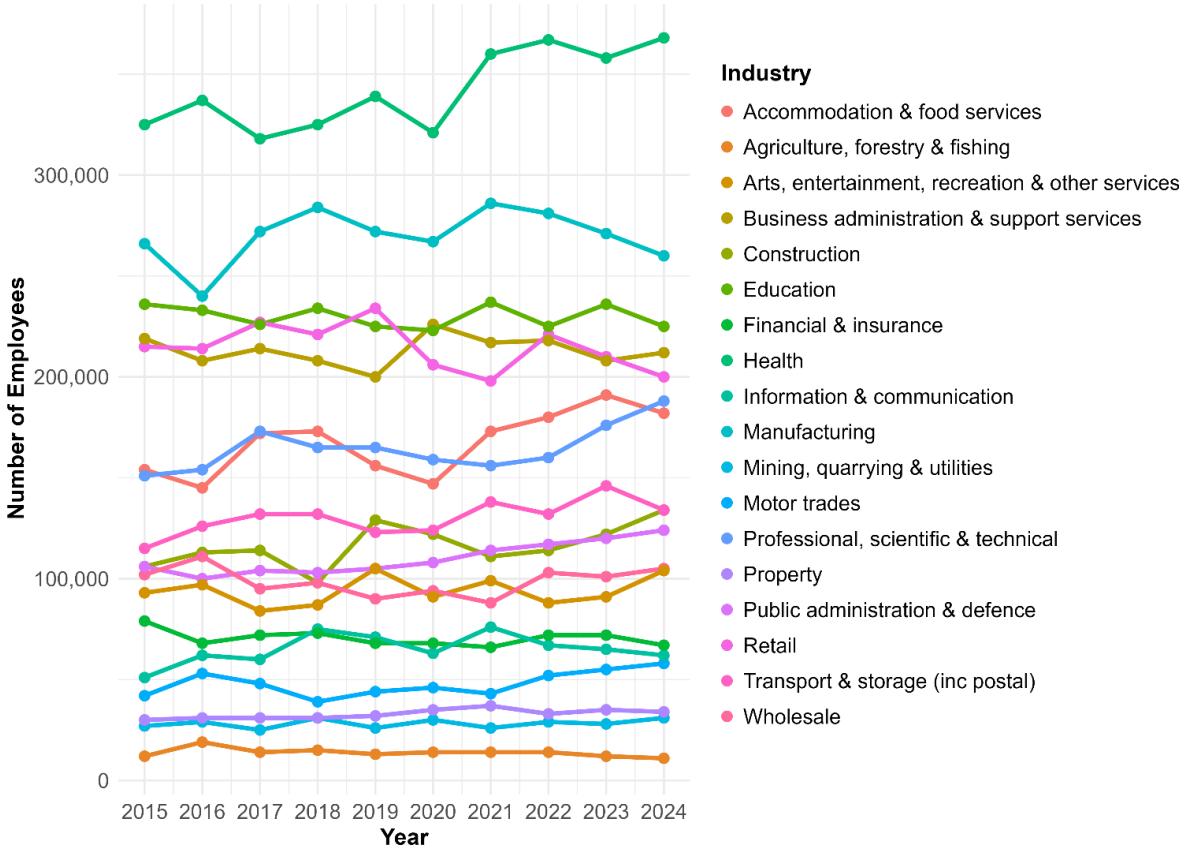
To address this, a composite visualisation, shown in Figure 1, presents multiple perspectives on the data. The analysis uses data from the Business Register and Employment Survey (BRES), an official annual survey of employee numbers. The dataset covers all Standard Industrial Classification sectors, providing annual employment counts per industry. This enables the identification of temporal trends, relative employment changes and pre-pandemic versus post-pandemic recovery patterns. Data preparation involved cleaning and standardising industry names, removing unnecessary flags and converting the dataset into long format to make visualisation and analysis easier. Attention is given to sectoral exposure to lockdown measures, remote working capacity and demand shocks. Service industries linked to hospitality and retail show sharp contractions during 2020, while health, logistics and public administration display stronger employment stability. Post-2021 trends highlight uneven recovery speeds, persistent labour reallocation and emerging structural shifts linked to digital adoption, hybrid work and regional policy support. These patterns inform interpretation of resilience, vulnerability and term regional adjustment.

### 2.1.1 Chart A: Line Chart - Employment Trends by Industry (2015-2024)

Figure 2 presents a line chart of employment trends across all industries in Yorkshire and the Humber. Plotting annual employee numbers reveals gradual shifts and abrupt changes due to the pandemic. This chart helps to identify industries with steady growth, those in decline and sectors most affected by pandemic-related shocks, providing essential context for understanding regional employment dynamics.

**Figure 2**

*Line Chart: Employment Trends by Industry (2015-2024)*



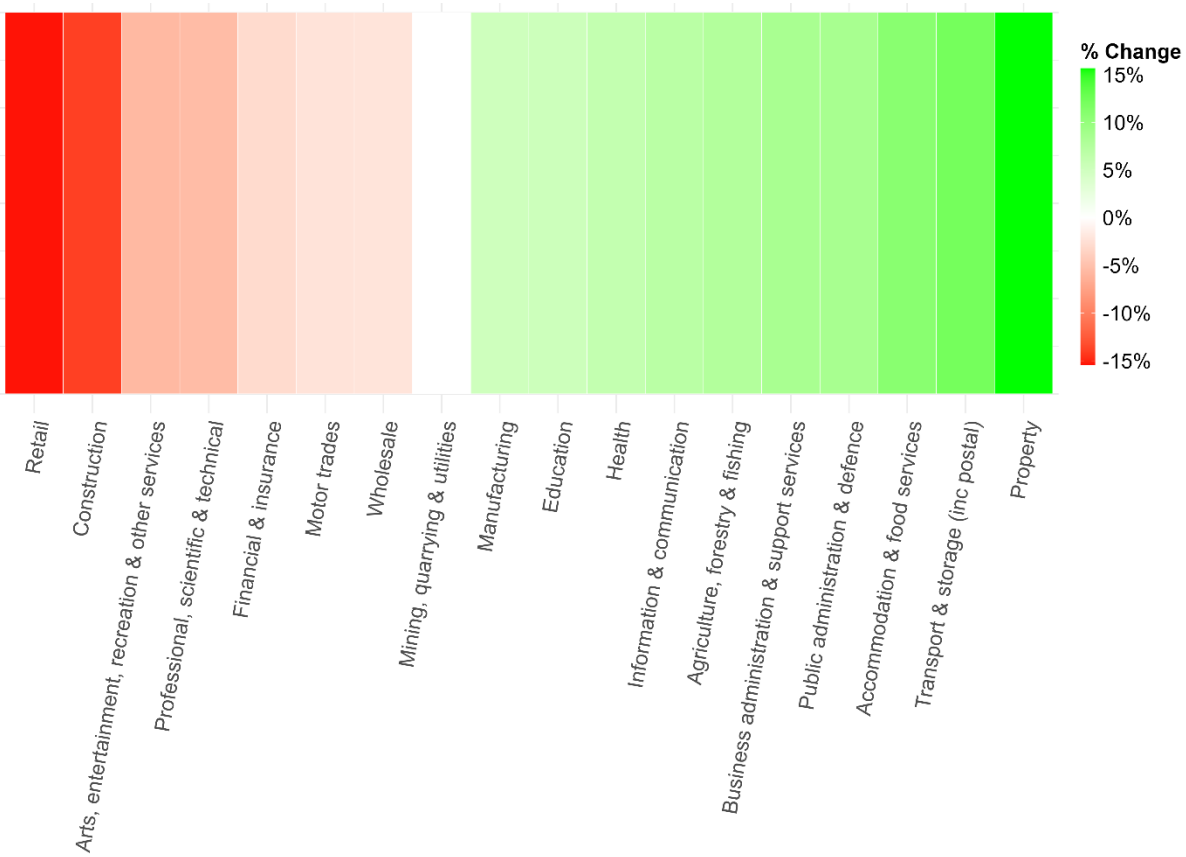

*Note*. This chart illustrates long-term employment trends and pandemic-related disruptions. Each industry is distinguished by a unique colour.

### 2.1.2 Chart B: Heatmap - Relative Employment Change (2019-2021)

Figure 3 uses a heatmap to illustrate the percentage change in employment between 2019 and 2021, capturing the pandemic's immediate impact (Hao et al., 2022). Positive changes are shown in green, negative changes in red and neutral values in white, allowing the industries experiencing the largest gains or losses to be rapidly identified. The diverging colour palette allows both magnitude of change and direction to be represented (Słomska-Przech et al., 2021).

**Figure 3**

*Heatmap: Relative Employment Change by Industry (2019-2021)*



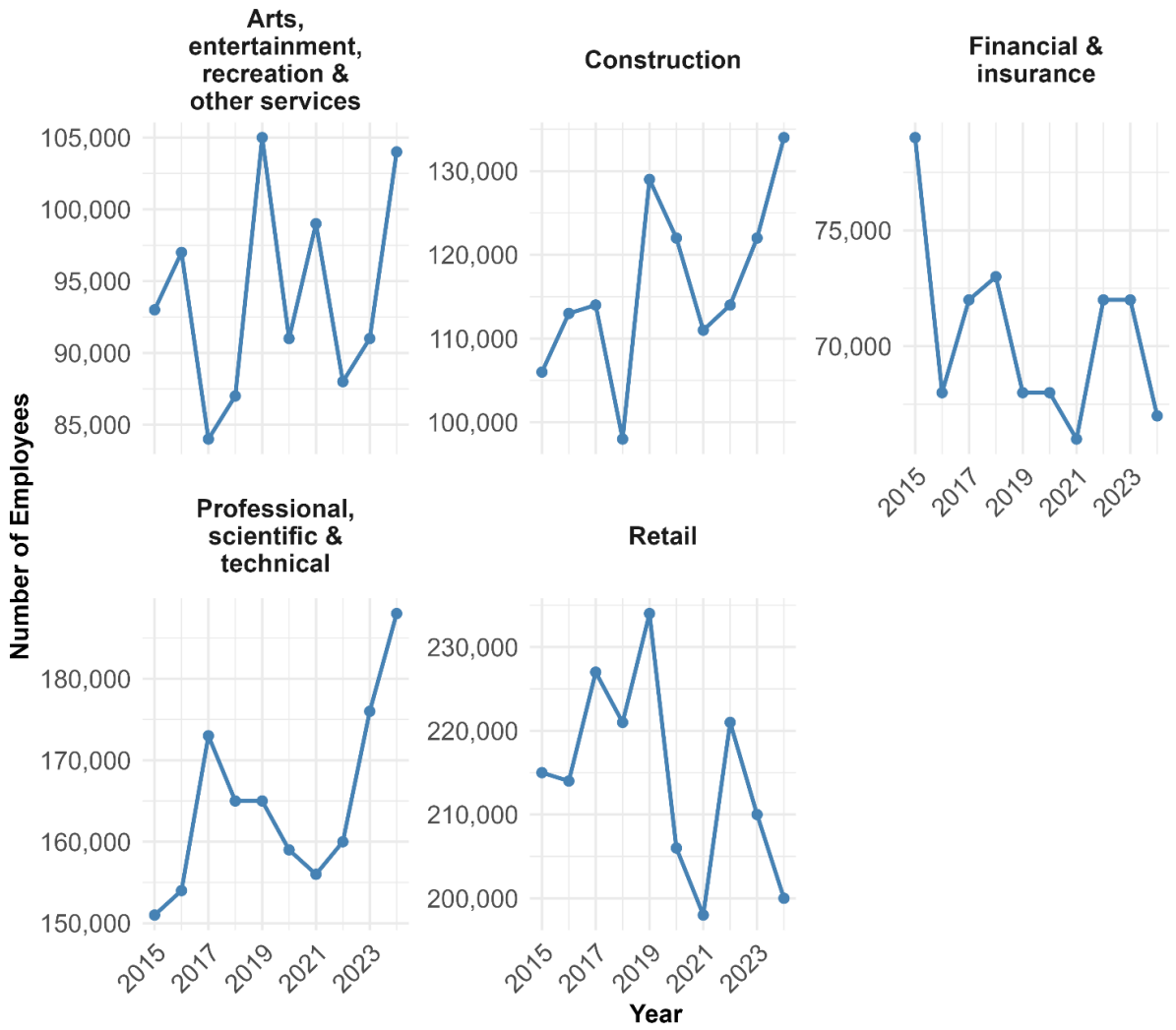

*Note*. Positive changes are shown in green, negative changes in red and neutral values in white, supporting quick identification of industries most affected by the pandemic.

### 2.1.3 Chart C: Small Multiples - Top 5 Most Affected Industries

Figure 4 presents small multiples focusing on the five industries most affected by the pandemic. Each panel shows annual employment trends for a single industry, making it easier to compare resilience across sectors. This format reduces visual clutter compared to overlapping lines and aligns with Tufte's (2001) principles of graphical excellence, highlighting structural vulnerabilities and the effectiveness of recovery policies.

**Figure 4**

*Small Multiples: Employment Trends for Top 5 Most Affected Industries*



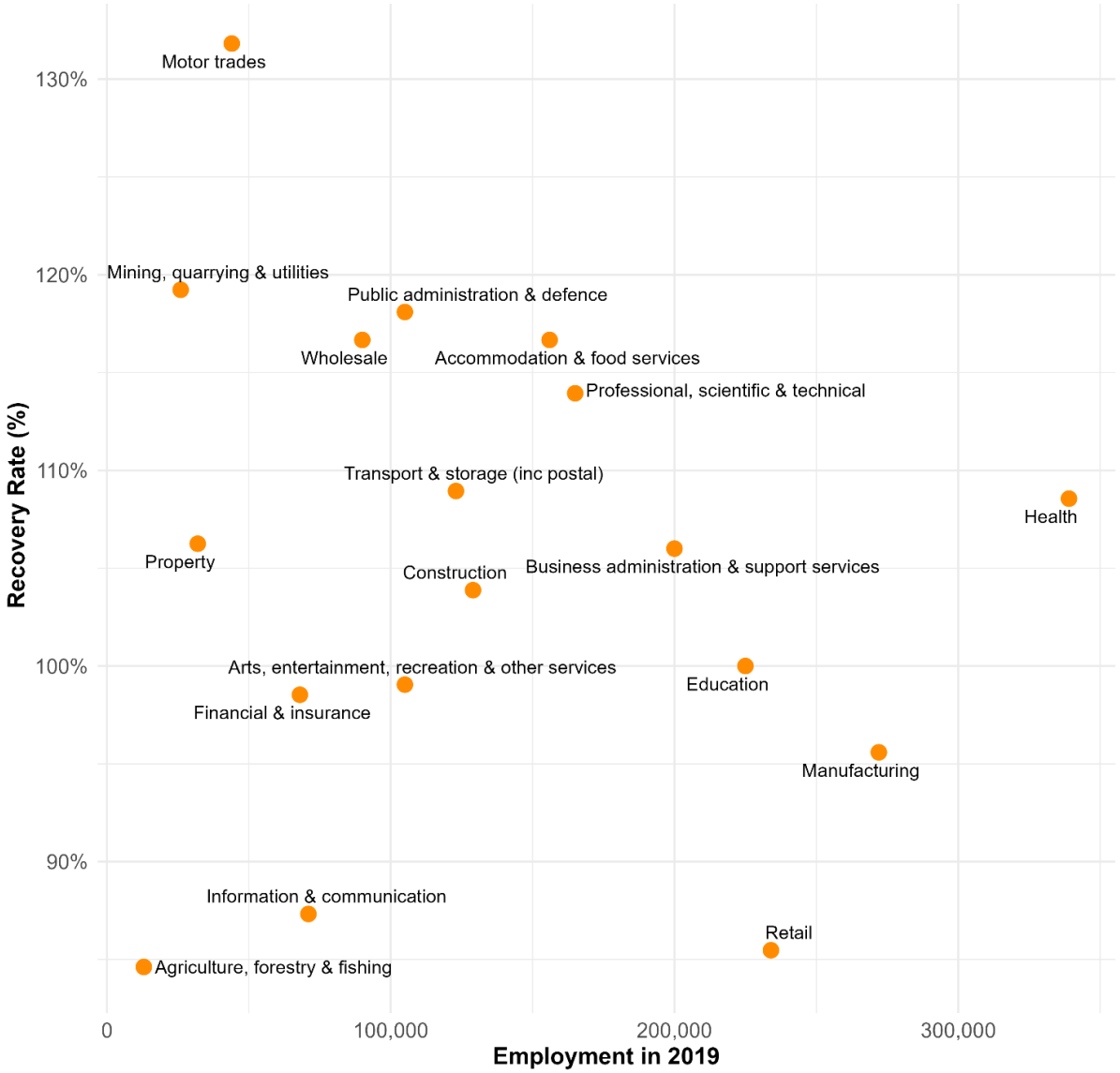

*Note.* Each panel displays the annual employment trend for one of the five industries with the largest employment changes, allowing comparison of recovery trajectories and identification of sectoral vulnerabilities.

### 2.1.4 Chart D: Scatterplot - Pre-Pandemic Employment Size vs Recovery Rate (2019/2024)

Figure 5 compares each industry's employment in 2019 with its recovery rate in 2024, calculated as the ratio of 2024 to pre-pandemic employment. This chart identifies industries that recovered quickly relative to their size and those lagging behind, complementing insights from Charts A-C. Such analysis is valuable for policymakers prioritising support for underperforming sectors (Office for National Statistics, 2022).

**Figure 5**

*Scatterplot: Pre-Pandemic Employment Size vs Recovery Rate (2019-2024)*




*Note*. The scatterplot compares each industry's employment in 2019 with its recovery rate in 2024, providing insight into relative sectoral resilience.

### 2.1.5 Integrated Knowledge from the Composite Visualisation

Together, these four charts provide a comprehensive picture of employment dynamics. Each chart offers a distinct perspective on temporal trends, relative change, detailed recovery and size versus resilience, while the composite visualisation allows the synthesis and cross-validation of insights. For example, industries with strong declines in the heatmap often correspond to those highlighted in the small multiples and scatterplot. This combination supports nuanced understanding of structural patterns that may not be apparent from individual charts alone (Lian et al., 2025).

### 2.1.6 Literature Context

The choice and combination of charts follow best practices in data visualisation. Small multiples and scatterplots are effective for comparing categorical trends and relationships (Nguyen et al., 2020). Diverging colour scales in heatmaps improve interpretability of gains and losses. Using multiple visual modalities aligns with cognitive theories of multimedia learning, which suggest that complementary visual channels improves knowledge building and retention (Mayer, 2002). This integrated approach allows complex regional employment patterns to be communicated clearly and efficiently.

## 2.2 THEORETICAL FRAMEWORKS

Chart A, the line chart representing employment trends across industries in Yorkshire (Figure 2), was developed using two major visualisation frameworks: ASSERT and the Grammar of Graphics (GoG). ASSERT provides a structured approach to designing effective visualisations, while GoG describes how individual graphical elements are combined to convey meaning (Wilkinson, 2005). Together, these frameworks make sure that the chart is theoretically grounded and technically coherent.

### 2.2.1 Applying the ASSERT Framework

The ASSERT framework (Ask, Structure, State, Engage, Refine, Tell) guides the conceptual and iterative development of a visualisation (Kelleher & Wagener, 2011).

#### *2.2.1.1 Ask*

The guiding question for the line chart was: "How have employment levels across industries in Yorkshire and the Humber changed between 2015 and 2024?" This focuses on long-term patterns, highlighting both growth and decline, and contextualising pandemic-related disruptions.

#### *2.2.1.2 Structure*

BRES data were reshaped into long format, with each row representing employment for a single industry in a given year. This structure aligns with GoG principles, where each observation maps directly to a visual element (Wilkinson, 2005). Structuring the data this way also supports scalability and consistency across years and industries.

#### *2.2.1.3 State*

The intended message of the chart is that industry employment trends vary across sectors. Some industries show resilience, others sustained decline. The multi-series design allows simultaneous comparison and highlights structural variability within the regional economy. This aligns with best practices in data storytelling (Segel & Heer, 2010).

#### *2.2.1.4 Engage*

User engagement is supported through the use of colour differentiation by industry and the annotation of key points around 2020-2021, the peak of the pandemic. Colour is a highly effective preattentive feature for categorical distinction, supporting rapid recognition without overwhelming the viewer (Ware, 2013). Engagement is further pushed by the ability to trace individual industries over time, encouraging exploratory analysis.

#### *2.2.1.5 Refine*

Refinement involved iterative adjustments to scales, labels and line thickness for clarity and accessibility. The x-axis uses discrete breaks for each year, the y-axis has formatted numerical labels for readability, and line colours were selected to ensure contrast for viewers with colour-vision deficiencies.

#### *2.2.1.6 Tell*

The line chart serves as the foundational "storyline" within the composite visualisation. It establishes the overall employment trajectory, providing context for the heatmap, small multiples and scatterplot. This approach follows Cairo's (2016) principle that effective visualisations guide audiences from overview to detail through coherent narrative progression.

### 2.2.2 Applying the Grammar of Graphics

The Grammar of Graphics conceptualises a visualisation as a layered system of components: data, aesthetics, geometries, scales, coordinates and facets (Wilkinson, 2005).

#### *2.2.2.1 Data*

The dataset contains industry, year and employment count variables, with each observation representing one industry-year pair.

#### *2.2.2.2 Aesthetics*

The x-aesthetic maps to Year, the y-aesthetic maps to Employees, and colour represents Industry, forming temporal and categorical dimensions. This combination supports trend comparison and group perception (Munzner, 2014).

#### *2.2.2.3 Geometries*

In R, geom_line connects data points to illustrate continuous trends, while geom_point emphasises individual years. This ensures both continuity and precise yearly values are visible.

#### *2.2.2.4 Scales*

A continuous x-scale is used for uniform temporal representation, while the y-scale uses comma-formatted numeric labels for readability. The colour scale encodes industry identity without implying magnitude.

#### *2.2.2.5 Coordinate System*

A Cartesian coordinate system with linear x and y-axes was used, suitable for displaying quantitative change over time (Cleveland & McGill, 1984).

#### *2.2.2.6 Theme and labels*

A minimal theme reduces non-data ink and emphasises the data lines themselves (Tufte, 2001). Titles, axis labels and legends use bold, high-contrast fonts for clarity. This GoG implementation balances precision and aesthetic simplicity, and allows reproducibility and modification, as each component can be adjusted independently without affecting the overall visual integrity.

### 2.2.3 Synthesis

Integrating ASSERT and GoG makes the line chart conceptually purposeful and technically coherent. ASSERT helps to formulate a systematic rationale grounded in audience needs and communication goals, and GoG provides the structural grammar to implement these intentions through data-aesthetic mappings. Together, they transform raw data into an interpretable narrative, making the line chart the analytical backbone of the composite visualisation.

## 2.3 ACCESSIBILITY

Accessibility in data visualisation means designing graphics so that they can be perceived, interpreted and acted upon by audiences with diverse perceptual and cognitive needs. Effective design supports users across the upper stages of the DIKW model, where visual representation helps transform data into information, knowledge and ultimately understanding (Ackoff, 1989). Chart B, the heatmap of relative employment change (Figure 3), was designed with these principles in mind to help users interpret the employment effects of the pandemic. This approach aligns with Kirk's Model of Understanding, which emphasises supporting perception, interpretation and comprehension (Kirk, 2016).

### 2.3.1 Accessibility Principles and Their Application

The heatmap uses a diverging colour palette to encode positive and negative percentage changes relative to a neutral midpoint. Diverging scales make it easy for users to distinguish direction and magnitude of change, which is important for interpreting employment disruption. As colour alone is not sufficient for accessibility, additional design features support interpretation. These include clear axis titles, full industry labels, a concise legend and an ordering of industries from the largest negative percentage change to the largest positive change, which reduces scanning effort and helps users follow the pattern more intuitively.

The heatmap also applies the design principle of abstraction, converting complex numerical changes into a structured grid of intensities. This helps users identify patterns without being overwhelmed by raw values. Reducing unnecessary visual complexity, while preserving the analytical message, supports sense making for users with varying levels of expertise.

For better perceptual accessibility, the colour palette was checked against common forms of colour vision deficiency. Although slight distinctions may remain challenging for a small proportion of viewers, labelled axes and a descriptive title provide an alternative route to interpretation for users who cannot rely solely on colour cues.

### 2.3.2 Potential Accessibility Limitations

Despite being designed for clarity, the heatmap has inherent limitations. Large categorical displays can increase cognitive load, making it difficult for users to maintain an overview while inspecting fine detail. The number of industries included risks visual crowding. Strategies such as grouping related sectors or using interactivity in a digital environment could help reduce this burden.

Colour encoding also presents potential barriers. Subtle gradients may be difficult for viewers with more severe colour vision limitations to distinguish (Emery & Webster, 2019). Redundant encoding, such as embedding numeric values within cells, could improve accessibility, although it may introduce clutter. Accessibility decisions must therefore balance clarity, simplicity and visual economy.

Interpretative demands are another consideration. Understanding the meaning of percentage change and the associated colour scale requires some familiarity with employment metrics. Providing explanatory text alongside the figure mitigates this by offering essential context without overloading the graphic itself.

### 2.3.3 Inclusive Design and Broader Accessibility

Accessibility extends beyond the visual itself to include supporting text and clear labelling. Descriptive titles and axis labels guide the reader through the visual and provide essential context, acting as interpretative scaffolding. High contrast colour choices, readable font sizes and sufficient spacing were also applied to maintain legibility across different screen sizes and printed formats.

### 2.3.4 Summary

This section has shown how accessibility principles shaped the design of the heatmap. Through abstraction, clear labelling, colour contrast and structured organisation, the visual supports users in progressing from data to understanding. Although there are still limitations, the design reflects widely recognised guidance that visualisations should be iterative, context-aware and inclusive, allowing audiences to interpret labour market patterns without unnecessary barriers.

## 2.4 VISUALISATION CHOICE

The selection of an appropriate visualisation is central to communicating insights effectively. Chart C, the small multiples of the top 5 most affected industries (Figure 4), was chosen to present detailed temporal employment trends for the sectors most disrupted by the COVID-19 pandemic. Small multiples, also known as trellis or lattice charts, repeat the same chart structure across subsets of the data, allowing straightforward comparison across categories (Tufte, 2001). In this case, each panel shows one industry with years on the x-axis and employee counts on the y-axis. This layout enables clear, side-by-side examination of recovery patterns and sector-specific volatility.

### 2.4.1 Rationale for Small Multiples

Small multiples were selected because they support pattern recognition and comparison without introducing visual clutter. Plotting the five industries on a single line chart would result in overlapping lines and colour confusion, especially given the dramatic shifts observed in some sectors. By placing each industry in its own panel, users can identify subtle differences in slopes, peaks and recovery timing with ease (Few, 2013). This aligns with the principle of chunking, which suggests that breaking information into manageable units improves comprehension (Miller, 1956). Consistent axes across panels promote comparability. Although y-axis scales vary to improve readability, the fixed x-axis allows users to visually align time points visually. This reflects a common design trade-off between absolute comparability and legibility (Heer et al., 2010).

### 2.4.2 Alternative Visualisation Options

Some alternatives were considered. An overlaid line chart with five colour coded series could have provided a global comparison but would likely introduce visual congestion and increase the risk of misinterpretation (Kelleher and Wagener, 2011). A stacked area chart was also an option, offering insight into cumulative employment across the selected industries while showing individual contributions. However, stacked areas tend to obscure individual trajectories, particularly for industries positioned lower in the stack. Compared with these alternatives, small multiples preserve the distinctness of each trend, making them more suitable for detailed comparative analysis.

### 2.4.3 Advantages of Small Multiples

Small multiples offer many different strengths for this dataset. Each panel is distinct, allowing focused inspection while retaining the ability to compare across industries. Subtle changes, turning points and differences in recovery speed are easier to detect than in aggregated or overlaid charts. The approach also scales well. Additional industries could be included without overwhelming the viewer, provided that the panel layout is adjusted. Evidence suggests small multiples are particularly effective for comparing categorical time series, aligning with established theories of visual perception (Nguyen et al., 2020).

### 2.4.4 Limitations and Considerations

Small multiples also have limitations. If too many categories are included, panels may become small and difficult to read, increasing the cognitive load on the user. This is addressed in the present design by restricting the chart to five industries. Varying y-axis scales can complicate direct numerical comparison but improve clarity within each panel. The technique also relies on users being familiar with repeated chart structures. Clear titles and labels help mitigate this by guiding interpretation.

### 2.4.5 Conclusion

The use of small multiples for Chart C reflects a deliberate balance between clarity, comparability and cognitive load. Although other visualisation methods give off different perspectives, the small multiples format most effectively communicates the detailed recovery trajectories of the industries most affected by the pandemic. By drawing on established visualisation literature, this design supports the broader goal of explaining employment changes in Yorkshire and the Humber during the post-pandemic period.

## 2.5 Implications and Improvements

Chart D, the scatterplot comparing pre-pandemic employment size in 2019 with the 2024 recovery rate (Figure 5), offers a clear overview of industry resilience after COVID-19. Although effective, there are ethical considerations that come with this type of visualisation, and interpretive challenges that must be addressed for responsible communication.

### 2.5.1 Ethical Considerations

Visualisations influence how audiences perceive data and can shape policy decisions. Because Chart D highlights recovery differences, it could be used to justify prioritising support for sectors that appear more resilient, which could potentially disadvantage industries with weaker recovery profiles (Few, 2013). Misinterpretation may also lead stakeholders to assume that larger industries are inherently more robust, even though recovery often depends on factors such as government support, regional policy and workforce composition (Office for National Statistics, 2022).

The chart also simplifies complex dynamics by presenting a single recovery metric without accounting for external influences including automation, supply chain pressures or shifts in consumer behaviour. Ethical use therefore requires clear annotation, a transparent explanation of methods and explicit recognition of contextual limitations to avoid encouraging oversimplified conclusions (Cairo, 2016).

### 2.5.2 Potential Misinterpretation

Several features of the scatterplot could lead to misunderstanding. The recovery rate is a relative measure. Without clear labelling, viewers may assume it represents absolute employment change, which could distort perceptions of sector performance. Industries also differ considerably in size and economic importance. Small industries with high relative recovery may appear more successful than large industries with slower rebounds, even if their overall impact differs. The scatterplot does not capture wider factors, such as policy interventions or market disruptions, which further complicate interpretation. These risks can be reduced through detailed captions, unambiguous axis labels and accompanying text that explains the meaning of the recovery ratio and highlights the need for contextual understanding.

### 2.5.3 Improvements and Enhancements

There are some refinements that could increase the clarity and accessibility of Chart D. Using bubble size to encode absolute 2019 employment would allow viewers to interpret relative recovery alongside industry scale, reducing the risk of overemphasising small sectors. Colour coding industries by broad economic grouping, such as primary, secondary and tertiary, would support pattern recognition and help contextualise trends.

In digital formats, interactive features such as tooltips displaying exact values, explanations or links to time series trends would enhance transparency. Targeted annotations can draw attention to outliers or sectors with notable disparities between size and recovery rate. Providing brief notes on relevant government interventions or economic strategies would further situate the chart within real world decision-making.

### 2.5.4 Conclusion

Chart D summarises industry recovery effectively but requires careful framing to prevent misinterpretation. Ethical visualisation practice demands transparency, acknowledgement of influencing factors and adequate contextual guidance. Enhancements such as size scaling, sector-based colour coding and annotation would improve clarity and reliability. These improvements align with established principles of visualisation ethics and support informed judgment by policymakers and stakeholders (Cairo, 2016; Kelleher and Wagener, 2011).

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# 4 APPENDIX

## 4.1 R CODE

```
#=========================================================#
# Yorkshire Industry Employment Analysis 2015-2024
# Composite Visualisation for IJC445
#=========================================================#


#-------------------------#
# Libraries
#-------------------------#
library(readxl)      # read Excel files
library(tidyverse)   # data manipulation & plotting
library(stringr)     # string cleaning
library(ggplot2)     # plotting
library(ggpubr)      # combine multiple plots
library(ggrepel)     # prevent overlapping labels
library(scales)      # formatting numbers


#-------------------------#
# Load Data
#-------------------------#
bres <- read_excel("bres_yorkshire_industry.xlsx")


#-------------------------#
# Data Cleaning
```

```
#-------------------------#
bres <- bres %>%
  select(-contains("Flags")) %>%  # remove unnecessary flag columns
  mutate(
    Industry = Industry %>%
      str_replace("^\\d+\\s*:\\s*", "") %>%  # remove numbering
      str_replace("\\s*\\([^)]*\\)$", "")    # remove trailing brackets
  )

# Convert to long format for easier plotting
bres_long <- bres %>%
  pivot_longer(cols = `2015`:`2024`, names_to = "Year", values_to =
"Employees") %>%
  mutate(Year = as.numeric(Year))

#-------------------------#
# Helper Function to Save Plots
#-------------------------#
save_plot <- function(plot, filename, width = 12, height = 12, dpi = 300){
  ggsave(filename, plot, width = width, height = height, dpi = dpi)
}

#-------------------------#
# Chart A / Figure 2: Line Chart - Employment Trends
#-------------------------#
```

```
line_chart <- ggplot(bres_long, aes(x = Year, y = Employees, colour =
Industry)) +
  geom_line(size = 2) +
  geom_point(size = 4) +
  scale_y_continuous(labels = comma) +
  scale_x_continuous(breaks = 2015:2024) +
  labs(title = "Employment Trends by Industry (2015-2024)",
       x = "Year",
       y = "Number of Employees",
       colour = "Industry") +
  theme_minimal(base_size = 20) +
  theme(
    plot.title = element_text(face = "bold", size = 26, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 22),
    axis.text = element_text(size = 20),
    legend.title = element_text(size = 22, face = "bold"),
    legend.text = element_text(size = 20),
    legend.key.height = unit(1.2, "cm"),
    legend.key.width = unit(0.25, "cm")
  ) +
  annotate("text", x = min(bres_long$Year),
           y = max(bres_long$Employees)*1.05,
           label = "A", size = 12, fontface = "bold", hjust = 0)

save_plot(line_chart, "line_chart.png", width = 16, height = 12)
```

```
#-------------------------#
# Chart B / Figure 3: Heatmap - Relative Employment Change
#-------------------------#
bres_change <- bres_long %>%
  filter(Year %in% c(2019, 2021)) %>%
  pivot_wider(names_from = Year, values_from = Employees) %>%
  mutate(Change_Percent = (`2021` - `2019`) / `2019` * 100) %>%
  arrange(Change_Percent) %>%
  mutate(Industry = factor(Industry, levels = Industry))

heatmap_chart <- ggplot(bres_change, aes(x = Industry, y = 1, fill =
Change_Percent)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(
    low = "red", mid = "white", high = "green", midpoint = 0,
    labels = scales::percent_format(scale = 1)
  ) +
  labs(
    title = "Relative Employment Change by Industry (2019-2021)",
    x = NULL,
    y = NULL,
    fill = "% Change"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    axis.text.y = element_blank(),
```

```
    axis.ticks.y = element_blank(),
    axis.text.x = element_text(angle = 80, hjust = 1, size = 20),
    plot.title = element_text(face = "bold", size = 23, hjust = 0.5),
    legend.title = element_text(size = 20, face = "bold"),
    legend.text = element_text(size = 20),
    legend.key.height = unit(2, "cm"),
    legend.key.width = unit(0.5, "cm")
  ) +
  annotate("text", x = 1, y = 1.2, label = "B", size = 12, fontface =
"bold", hjust = 0)

save_plot(heatmap_chart, "heatmap_chart.png", width = 16, height = 12)

#-------------------------#
# Chart C / Figure 4: Small Multiples - Top 5 Industries
#-------------------------#
top5_industries <- bres_change %>%
  slice(1:5) %>%
  pull(Industry)

bres_top5 <- bres_long %>% filter(Industry %in% top5_industries)

small_multiples <- ggplot(bres_top5, aes(x = Year, y = Employees)) +
  geom_line(colour = "steelblue", size = 1.5) +
  geom_point(colour = "steelblue", size = 3) +
```

```
  facet_wrap(~Industry, scales = "free_y", labeller = label_wrap_gen(width
= 15)) +
  scale_y_continuous(labels = comma) +
  scale_x_continuous(breaks = seq(2015, 2024, 2)) +
  labs(title = "Employment Trends for Top 5 Most Affected Industries",
       x = "Year",
       y = "Number of Employees") +
  theme_minimal(base_size = 20) +
  theme(
    plot.title = element_text(face = "bold", size = 22, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 20),
    axis.text = element_text(size = 20),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(face = "bold", size = 20),
    panel.spacing = unit(1, "lines")
  ) +
  annotate("text", x = min(bres_top5$Year),
           y = max(bres_top5$Employees)*1.05,
           label = "C", size = 12, fontface = "bold", hjust = 0)

save_plot(small_multiples, "small_multiples.png", width = 13, height = 12)

#-------------------------#
# Chart D / Figure 5: Scatterplot - Pre-Pandemic Size vs Recovery Rate
#-------------------------#
bres_recovery <- bres_long %>%
```

```
  filter(Year %in% c(2019, 2024)) %>%
  pivot_wider(names_from = Year, values_from = Employees) %>%
  mutate(Recovery_Rate = `2024` / `2019`)

scatter_chart <- ggplot(bres_recovery, aes(x = `2019`, y = Recovery_Rate,
label = Industry)) +
  geom_point(colour = "darkorange", size = 5) +
  geom_text_repel(size = 5, max.overlaps = 20, box.padding = 0.5,
point.padding = 0.5, segment.color = "grey50") +
  scale_x_continuous(labels = comma) +
  scale_y_continuous(labels = percent_format(scale = 100)) +
  labs(title = "Pre-Pandemic Employment Size vs Recovery Rate
(2024/2019)",
       x = "Employment in 2019",
       y = "Recovery Rate (%)") +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 22, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 18),
    axis.text = element_text(size = 16)
  ) +
  annotate("text", x = min(bres_recovery$`2019`),
           y = max(bres_recovery$Recovery_Rate)*1.05,
           label = "D", size = 12, fontface = "bold", hjust = 0)

save_plot(scatter_chart, "scatter_chart.png", width = 12, height = 12)
```

```
#------------------------#
# Composite Visualisation
#------------------------#
composite_plot <- ggarrange(
  line_chart, heatmap_chart, small_multiples, scatter_chart,
  ncol = 2, nrow = 2,
  common.legend = FALSE
)

save_plot(composite_plot, "composite_visualisation.png", width = 32,
height = 24)

#=========================================================#
# End of Script
#=========================================================#
```

## 4.2 GitHub Page

All R code, the dataset, and generated visualisations for this project are available on GitHub at [GitHub repository]. The repository (yorkshire_employment_visualisation) includes a fully commented R script (employment_visualisation_code.R), the dataset of annual employee counts by industry (bres_yorkshire_industry.xlsx), and a folder of all plots, including the line chart, heatmap, small multiples, scatterplot and the composite visualisation. The repository also contains a README file summarising the project overview, key research questions, visualisation approach and main findings.